

Systems of Linear Equations

Solving systems of equations using matrices

A system of linear equations is a set of n equations in n unknowns (variables) of the form

$$\begin{aligned}a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\&\vdots \\a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n &= b_n\end{aligned}$$

where, the unknowns are denoted by $x_1, x_2, \dots, x_n$, the coefficients (a_{ij}) and constants (b_i) are given values. The system of equations above can be written in matrix form as

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \quad \text{or} \quad \mathbf{Ax} = \mathbf{b}$$

where, $\mathbf{A}$ is matrix of the coefficients (a_{ij}) , $\mathbf{x}$ is a vector of n unknowns and $\mathbf{b}$ is a vector of n constants.

A **solution** to linear system (1) is a sequence of n numbers $s_1, s_2, \dots, s_n$, which has the property that each equation in (2) is satisfied when $x_1 = s_1, x_2 = s_2, \dots, x_n = s_n$ are substituted.

If the linear system (1) has no solution, it is said to be **inconsistent**; if it has a solution, it is called **consistent**. If $b_1 = b_2 = \dots = b_m = 0$, then (1) is called a **homogeneous system**. Note that $x_1 = x_2 = \dots = x_n = 0$ is always a solution to a homogeneous system; it is called the **trivial solution**. A solution to a homogeneous system in which not all of $x_1, x_2, \dots, x_n$ are zero is called a **nontrivial solution**.

The director of a trust fund has \$100,000 to invest. The rules of the trust state that both a certificate of deposit (CD) and a long-term bond must be used. The director's goal is to have the trust yield \$7800 on its investments for the year. The CD chosen returns 5% per annum, and the bond 9%. The director determines the amount x to invest in the CD and the amount y to invest in the bond as follows:

Since the total investment is \$100,000, we must have $x + y = 100,000$. Since the desired return is \$7800, we obtain the equation $0.05x + 0.09y = 7800$. Thus, we have the linear system

$$\begin{aligned}x + y &= 100,000 \\0.05x + 0.09y &= 7800.\end{aligned}\tag{6}$$

To eliminate x , we add (-0.05) times the first equation to the second, obtaining

$$0.04y = 2800,$$

an equation having no x term. We have eliminated the unknown x . Then solving for y , we have

$$y = 70,000,$$

and substituting into the first equation of (6), we obtain

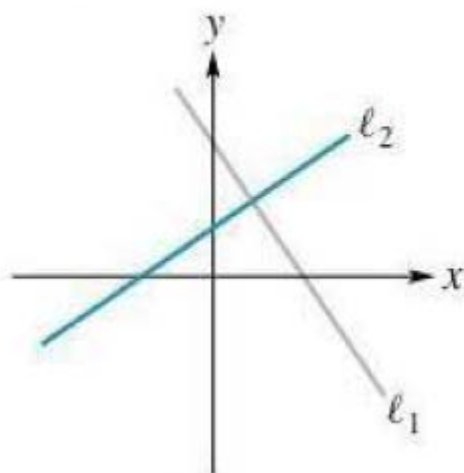
$$x = 30,000.$$

Consider next a linear system of two equations in the unknowns x and y :

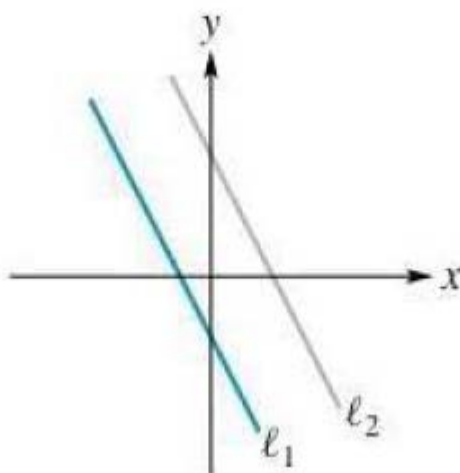
$$a_1x + a_2y = c_1$$

$$b_1x + b_2y = c_2.$$

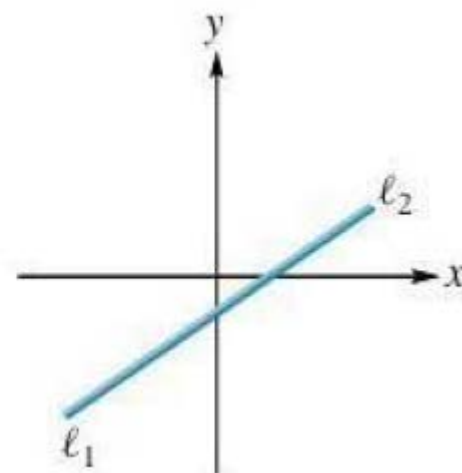
(a) A unique solution.



(b) No solution.



(c) Infinitely many solutions.



Next, consider a linear system of three equations in the unknowns x , y , and z :

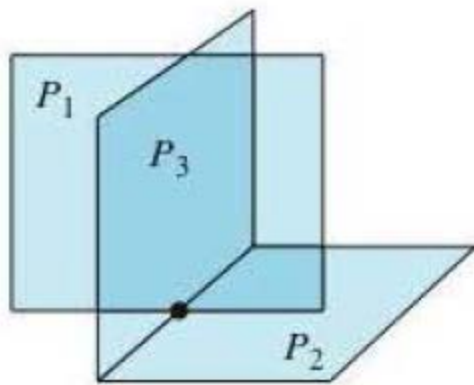
$$a_1x + b_1y + c_1z = d_1$$

$$a_2x + b_2y + c_2z = d_2$$

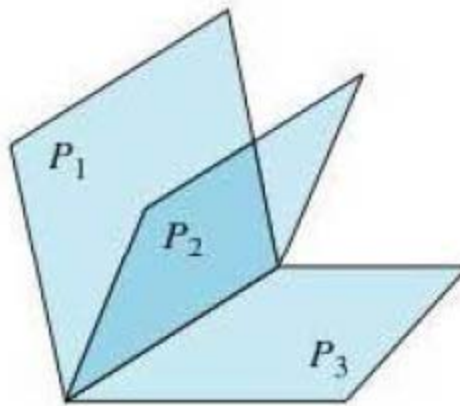
$$a_3x + b_3y + c_3z = d_3.$$

The graph of each of these equations is a plane, denoted by P_1 , P_2 , and P_3 , respectively. As in the case of a linear system of two equations in two unknowns,

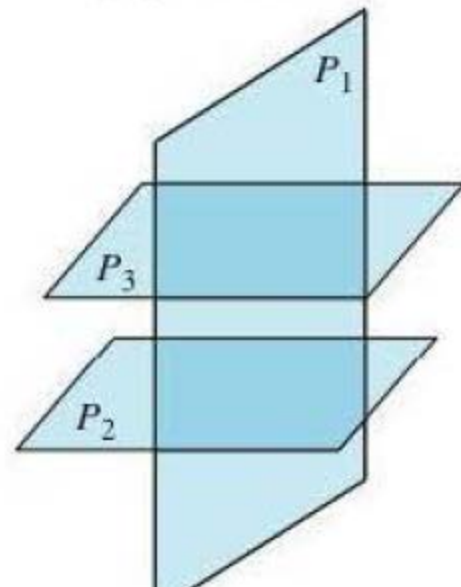
(a) A unique solution.



(b) Infinitely many solutions.



(c) No solution.



Consider the linear system of m equations in n unknowns,

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$

$$\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m.$$

Now define the following matrices:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}.$$

Then

$$\begin{aligned} A\mathbf{x} &= \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \\ &= \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{bmatrix}. \end{aligned}$$

$$A\mathbf{x} = \mathbf{b}.$$

The matrix A is called the **coefficient matrix** of the linear system, and the matrix

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array} \right],$$

obtained by adjoining column $\mathbf{b}$ to A , is called the **augmented matrix** of the linear system. The augmented matrix of () is written as $[A \mid \mathbf{b}]$. Conversely, any matrix with more than one column can be thought of as the augmented matrix of a linear system. The coefficient and augmented matrices play key roles in our method for solving linear systems.

$$b_1 = b_2 = \cdots = b_m = 0$$

in (4), the linear system is called a **homogeneous system**. A homogeneous system can be written as

$$A\mathbf{x} = \mathbf{0},$$

where A is the coefficient matrix.

Consider the linear system

$$\begin{aligned}-2x \quad \quad + z &= 5 \\ 2x + 3y - 4z &= 7 \\ 3x + 2y + 2z &= 3.\end{aligned}$$

Letting

$$A = \begin{bmatrix} -2 & 0 & 1 \\ 2 & 3 & -4 \\ 3 & 2 & 2 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 5 \\ 7 \\ 3 \end{bmatrix},$$

we can write the given linear system in matrix form as

$$A\mathbf{x} = \mathbf{b}.$$

The coefficient matrix is A , and the augmented matrix is

$$\left[\begin{array}{ccc|c} -2 & 0 & 1 & 5 \\ 2 & 3 & -4 & 7 \\ 3 & 2 & 2 & 3 \end{array} \right].$$

To see how a linear system whose augmented matrix has a particular form can be readily solved, suppose that

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & -1 \end{array} \right]$$

represents the augmented matrix of a linear system. Then the solution is quickly

found from the corresponding equations

$$x_1 + 2x_2 = 3$$

$$x_2 + x_3 = 2$$

$$x_3 = -1$$

as

$$x_3 = -1$$

$$x_2 = 2 - x_3 = 2 + 1 = 3$$

$$x_1 = 3 - 2x_2 = 3 - 6 = -3.$$

The task of this section is to manipulate the augmented matrix representing a given linear system into a form from which the solution can be found more easily.

We now apply row operations to the solution of linear systems.

Theorem

Let $A\mathbf{x} = \mathbf{b}$ and $C\mathbf{x} = \mathbf{d}$ be two linear systems, each of m equations in n unknowns. If the augmented matrices $[A \mid \mathbf{b}]$ and $[C \mid \mathbf{d}]$ are row equivalent, then the linear systems are equivalent; that is, they have exactly the same solutions.

$$\begin{array}{cccc}
 a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n & = & 0 \\
 a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n & = & 0 \\
 \vdots & & \vdots \\
 a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n & = & 0
 \end{array} \tag{1}$$

is called a homogeneous system. We can also write (1) in matrix form as

$$A\mathbf{x} = \mathbf{0}. \tag{2}$$

If A and C are row equivalent $m \times n$ matrices, then the homogeneous systems $A\mathbf{x} = \mathbf{0}$ and $C\mathbf{x} = \mathbf{0}$ are equivalent.

We observe that we have developed the essential features of two very straightforward methods for solving linear systems. The idea consists of starting with the linear system $A\mathbf{x} = \mathbf{b}$, then obtaining a partitioned matrix $[C \mid \mathbf{d}]$ in either row echelon form or reduced row echelon form that is row equivalent to the augmented matrix $[A \mid \mathbf{b}]$. Now $[C \mid \mathbf{d}]$ represents the linear system $C\mathbf{x} = \mathbf{d}$, which is quite simple to solve because of the structure of $[C \mid \mathbf{d}]$, and the set of solutions to this system gives precisely the set of solutions to $A\mathbf{x} = \mathbf{b}$; that is, the linear systems $A\mathbf{x} = \mathbf{b}$ and $C\mathbf{x} = \mathbf{d}$ are equivalent. The method where $[C \mid \mathbf{d}]$ is in row echelon form is called **Gaussian elimination**; the method where $[C \mid \mathbf{d}]$ is in reduced row echelon form is called **Gauss*–Jordan[†] reduction**.



Gaussian elimination consists of two steps:

Step 1. The transformation of the augmented matrix $[A \mid \mathbf{b}]$ to the matrix $[C \mid \mathbf{d}]$ in row echelon form using elementary row operations

Step 2. Solution of the linear system corresponding to the augmented matrix $[C \mid \mathbf{d}]$ using **back substitution**

For the case in which A is $n \times n$, and the linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution, the matrix $[C \mid \mathbf{d}]$ has the following form:

$$\left[\begin{array}{cccccc|c} 1 & c_{12} & c_{13} & \cdots & & c_{1n} & d_1 \\ 0 & 1 & c_{23} & \cdots & & c_{2n} & d_2 \\ \vdots & \vdots & \vdots & & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & c_{n-1n} & d_{n-1} \\ 0 & 0 & 0 & \cdots & 0 & 1 & d_n \end{array} \right].$$

$$\begin{array}{rcl}
x_1 + c_{12}x_2 + c_{13}x_3 + \dots + c_{1n}x_n & = & d_1 \\
x_2 + c_{23}x_3 + \dots + c_{2n}x_n & = & d_2 \\
\vdots & & \vdots \\
x_{n-1} + c_{n-1n}x_n & = & d_{n-1} \\
x_n & = & d_n.
\end{array}$$

Back substitution proceeds from the n th equation upward, solving for one variable from each equation:

$$\begin{array}{l}
x_n = d_n \\
x_{n-1} = d_{n-1} - c_{n-1n}x_n \\
\vdots \\
x_2 = d_2 - c_{23}x_3 - c_{24}x_4 - \dots - c_{2n}x_n \\
x_1 = d_1 - c_{12}x_2 - c_{13}x_3 - \dots - c_{1n}x_n.
\end{array}$$

EXAMPLE

The linear system

$$x + 2y + 3z = 9$$

$$2x - y + z = 8$$

$$3x \quad \quad - z = 3$$

has the augmented matrix

$$[A \mid \mathbf{b}] = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 9 \\ 2 & -1 & 1 & 8 \\ 3 & 0 & -1 & 3 \end{array} \right].$$

Transforming this matrix to row echelon form, we obtain (verify)

$$[C \mid \mathbf{d}] = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 9 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right].$$

Using back substitution, we now have

$$z = 3$$

$$y = 2 - z = 2 - 3 = -1$$

$$x = 9 - 2y - 3z = 9 + 2 - 9 = 2;$$

thus the solution is $x = 2$, $y = -1$, $z = 3$, which is unique.

The general case in which A is $m \times n$ is handled in a similar fashion, but we need to elaborate upon several situations that can occur. We thus consider $C\mathbf{x} = \mathbf{d}$, where C is $m \times n$, and $[C \mid \mathbf{d}]$ is in row echelon form. Then, for example, $[C \mid \mathbf{d}]$ might be of the following form:

$$\left[\begin{array}{ccccccccc|c} 1 & c_{12} & c_{13} & \cdots & & & c_{1n} & d_1 \\ 0 & 0 & 1 & c_{24} & \cdots & & c_{2n} & d_2 \\ \vdots & \vdots & \vdots & \vdots & & & \vdots & \vdots \\ 0 & 0 & \cdots & & 0 & 1 & c_{k-1n} & d_{k-1} \\ 0 & \cdots & & & \vdots & 0 & 1 & d_k \\ 0 & \cdots & & & & \vdots & 0 & d_{k+1} \\ \vdots & & & & & & \vdots & \vdots \\ 0 & \cdots & & & & & 0 & d_m \end{array} \right].$$

This augmented matrix represents the linear system

$$\begin{array}{ccccccc}
 x_1 + c_{12}x_2 + c_{13}x_3 + \cdots & + & c_{1n}x_n & = & d_1 \\
 & & x_3 + c_{24}x_4 + \cdots + & c_{2n}x_n & = & d_2 \\
 & & \vdots & & & \\
 & & & & x_{n-1} + c_{k-1n}x_n & = & d_{k-1} \\
 & & & & x_n & = & d_k \\
 0x_1 + & + & \cdots & + & 0x_n & = & d_{k+1} \\
 \vdots & & \vdots & & \vdots & & \\
 0x_1 + & + & \cdots & + & 0x_n & = & d_m.
 \end{array}$$

First, if $d_{k+1} = 1$, then $C\mathbf{x} = \mathbf{d}$ has no solution, since at least one equation is not satisfied. If $d_{k+1} = 0$, which implies that $d_{k+2} = \cdots = d_m = 0$ (since $[C \mid \mathbf{d}]$ was assumed to be in row echelon form), we then obtain $x_n = d_k$, $x_{n-1} = d_{k-1} - c_{k-1n}x_n = d_{k-1} - c_{k-1n}d_k$ and continue using back substitution to find the remaining unknowns corresponding to the leading entry in each row. Of course, in the solution some of the unknowns may be expressed in terms of others that can take on any values whatsoever. This merely indicates that $C\mathbf{x} = \mathbf{d}$ has infinitely many solutions. On the other hand, every unknown may have a determined value, indicating that the solution is unique.

EXAMPLE

Let

$$[C \mid \mathbf{d}] = \left[\begin{array}{ccccc|c} 1 & 2 & 3 & 4 & 5 & 6 \\ 0 & 1 & 2 & 3 & -1 & 7 \\ 0 & 0 & 1 & 2 & 3 & 7 \\ 0 & 0 & 0 & 1 & 2 & 9 \end{array} \right].$$

Then

$$x_4 = 9 - 2x_5$$

$$x_3 = 7 - 2x_4 - 3x_5 = 7 - 2(9 - 2x_5) - 3x_5 = -11 + x_5$$

$$x_2 = 7 - 2x_3 - 3x_4 + x_5 = 2 + 5x_5$$

$$x_1 = 6 - 2x_2 - 3x_3 - 4x_4 - 5x_5 = -1 - 10x_5$$

$$x_5 = \text{any real number.}$$

The system is consistent, and all solutions are of the form

$$x_1 = -1 - 10r$$

$$x_2 = 2 + 5r$$

$$x_3 = -11 + r$$

$$x_4 = 9 - 2r$$

$$x_5 = r, \text{ any real number.}$$

Since r can be assigned any real number, the given linear system has infinitely many solutions. ■

If

$$[C \mid \mathbf{d}] = \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 5 \\ 0 & 1 & 0 & 0 & 6 \\ 0 & 0 & 1 & 0 & 7 \\ 0 & 0 & 0 & 1 & 8 \end{array} \right],$$

then the unique solution is

$$x_1 = 5$$

$$x_2 = 6$$

$$x_3 = 7$$

$$x_4 = 8.$$

If

$$[C \mid \mathbf{d}] = \left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 5 \\ 0 & 1 & 2 & 3 & 6 \\ 0 & 0 & 0 & 0 & 1 \end{array} \right],$$

then $C\mathbf{x} = \mathbf{d}$ has no solution, since the last equation is

$$0x_1 + 0x_2 + 0x_3 + 0x_4 = 1,$$

which can never be satisfied. ■

If

$$[C \mid \mathbf{d}] = \left[\begin{array}{ccccc|c} 1 & 1 & 2 & 0 & -\frac{5}{2} & \frac{2}{3} \\ 0 & 0 & 0 & 1 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right],$$

then

$$x_4 = \frac{1}{2} - \frac{1}{2}x_5$$

$$x_1 = \frac{2}{3} - x_2 - 2x_3 + \frac{5}{2}x_5,$$

where x_2 , x_3 , and x_5 can take on any real numbers, so the system has infinitely many solutions. Thus a solution is of the form

$$x_1 = \frac{2}{3} - r - 2s + \frac{5}{2}t$$

$$x_2 = r$$

$$x_3 = s$$

$$x_4 = \frac{1}{2} - \frac{1}{2}t$$

$$x_5 = t,$$

where r , s , and t are any real numbers. ■

Consider the linear system

$$\begin{aligned}x + 2y + 3z &= 6 \\2x - 3y + 2z &= 14 \\3x + y - z &= -2.\end{aligned}$$

■ Homogeneous Systems

Now we study a homogeneous system $A\mathbf{x} = \mathbf{0}$ of m linear equations in n unknowns.

Consider the homogeneous system whose augmented matrix is

$$\left[\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 1 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

Since the augmented matrix is in reduced row echelon form, the solution is seen to be

$$x_1 = -2r$$

$$x_2 = s$$

$$x_3 = -3r$$

$$x_4 = -4r$$

$$x_5 = r,$$

where r and s are any real numbers. ■

Theorem

A homogeneous system of m linear equations in n unknowns always has a non-trivial solution if $m < n$, that is, if the number of unknowns exceeds the number of equations.

We shall use this result in the following equivalent form: If A is $m \times n$ and $A\mathbf{x} = \mathbf{0}$ has only the trivial solution, then $m \geq n$.

EXAMPLE

Consider the homogeneous system

$$x + y + z + w = 0$$

$$x + w = 0$$

$$x + 2y + z = 0.$$

EXAMPLE

Consider the homogeneous system

$$x + y + z + w = 0$$

$$x + w = 0$$

$$x + 2y + z = 0.$$

The augmented matrix

$$A = \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 2 & 1 & 0 & 0 \end{array} \right]$$

The augmented matrix

$$A = \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 2 & 1 & 0 & 0 \end{array} \right]$$

is row equivalent to (verify)

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{array} \right].$$

Hence the solution is

$$x = -r$$

$$y = r$$

$$z = -r$$

$$w = r, \text{ any real number.}$$

Inverse matrix method

The matrix equation $\mathbf{Ax} = \mathbf{b}$, where $\mathbf{A}$ is an $n \times n$ regular matrix ($\det(\mathbf{A})$ is not 0), multiplied by $\mathbf{A}^{-1}$ gives

$$\mathbf{A}^{-1}\mathbf{Ax} = \mathbf{A}^{-1}\mathbf{b}$$

since, $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$ by definition and $\mathbf{Ix} = \mathbf{x}$, where $\mathbf{I}$ is the identity matrix, as

$$\mathbf{I} \cdot \mathbf{x} = \mathbf{x} \quad \Leftrightarrow \quad \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} 1 \cdot x_1 + 0 \cdot x_2 + \cdots + 0 \cdot x_n \\ 0 \cdot x_1 + 1 \cdot x_2 + \cdots + 0 \cdot x_n \\ \vdots \\ 0 \cdot x_1 + 0 \cdot x_2 + \cdots + 1 \cdot x_n \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

then

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}.$$

Thus, the inverse matrix method uses this matrix equation to find the solution to the system of equations directly.

Example: Find the solution of the given three equations using the inverse matrix method.

$$\begin{array}{rcl} 2x_1 + x_2 - x_3 = 1 \\ x_1 + 2x_2 + x_3 = 8 \\ \underline{3x_1 - x_2 - x_3 = -2} \end{array} \Leftrightarrow \begin{bmatrix} 2 & 1 & -1 \\ 1 & 2 & 1 \\ 3 & -1 & -1 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 8 \\ -2 \end{bmatrix} \equiv \mathbf{Ax} = \mathbf{b}.$$

Given equations are written as the equivalent matrix equation.

Then, both sides of the above matrix equation we multiply by the inverse of the coefficient matrix $\mathbf{A}$ (calculation of which is shown in the previous example),

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \cdot [A_{ji}], \quad \mathbf{A}^{-1} = \begin{bmatrix} -\frac{1}{9} & \frac{2}{9} & \frac{1}{3} \\ \frac{4}{9} & \frac{1}{9} & -\frac{1}{3} \\ -\frac{7}{9} & \frac{5}{9} & \frac{1}{3} \end{bmatrix}.$$

Since $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$ and $\mathbf{Ix} = \mathbf{x}$, then

$$\begin{aligned} \mathbf{x} = \mathbf{A}^{-1}\mathbf{b} &\Leftrightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -\frac{1}{9} & \frac{2}{9} & \frac{1}{3} \\ \frac{4}{9} & \frac{1}{9} & -\frac{1}{3} \\ -\frac{7}{9} & \frac{5}{9} & \frac{1}{3} \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 8 \\ -2 \end{bmatrix}, \\ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} &= \begin{bmatrix} -\frac{1}{9} + \frac{16}{9} - \frac{2}{3} \\ \frac{4}{9} + \frac{8}{9} - \frac{2}{3} \\ -\frac{7}{9} + \frac{40}{9} - \frac{2}{3} \end{bmatrix}, \quad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}. \end{aligned}$$

► **EXAMPLE 1** Solution of a Linear System Using A^{-1}

Consider the system of linear equations

$$\begin{aligned}x_1 + 2x_2 + 3x_3 &= 5 \\2x_1 + 5x_2 + 3x_3 &= 3 \\x_1 \quad \quad + 8x_3 &= 17\end{aligned}$$

In matrix form this system can be written as $A\mathbf{x} = \mathbf{b}$, where

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 5 \\ 3 \\ 17 \end{bmatrix}$$

In Example 4 of the preceding section, we showed that A is invertible and

$$A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$$

By Theorem 1.6.2, the solution of the system is

$$\mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \\ 17 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

or $x_1 = 1, x_2 = -1, x_3 = 2$. ◀

Cramer's* Rule

Let

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n = b_n$$

be a linear system of n equations in n unknowns, and let $A = [a_{ij}]$ be the coefficient matrix so that we can write the given system as $A\mathbf{x} = \mathbf{b}$, where

$$\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}.$$

If $\det(A) \neq 0$, then the system has the unique solution

$$x_1 = \frac{\det(A_1)}{\det(A)}, \quad x_2 = \frac{\det(A_2)}{\det(A)}, \quad \dots, \quad x_n = \frac{\det(A_n)}{\det(A)},$$

where A_i is the matrix obtained from A by replacing the i th column of A by $\mathbf{b}$.

Consider the following linear system:

$$-2x_1 + 3x_2 - x_3 = 1$$

$$x_1 + 2x_2 - x_3 = 4$$

$$-2x_1 - x_2 + x_3 = -3.$$

We have $|A| = \begin{vmatrix} -2 & 3 & -1 \\ 1 & 2 & -1 \\ -2 & -1 & 1 \end{vmatrix} = -2$. Then

$$x_1 = \frac{\begin{vmatrix} 1 & 3 & -1 \\ 4 & 2 & -1 \\ -3 & -1 & 1 \end{vmatrix}}{|A|} = \frac{-4}{-2} = 2,$$

$$x_2 = \frac{\begin{vmatrix} -2 & 1 & -1 \\ 1 & 4 & -1 \\ -2 & -3 & 1 \end{vmatrix}}{|A|} = \frac{-6}{-2} = 3,$$

and

$$x_3 = \frac{\begin{vmatrix} -2 & 3 & 1 \\ 1 & 2 & 4 \\ -2 & -1 & -3 \end{vmatrix}}{|A|} = \frac{-8}{-2} = 4.$$

We note that Cramer's rule is applicable only to the case in which we have n equations in n unknowns and the coefficient matrix A is nonsingular. If we have to solve a linear system of n equations in n unknowns whose coefficient matrix is singular, then we must use the Gaussian elimination or Gauss–Jordan reduction methods as discussed in [1]. Cramer's rule becomes computationally inefficient for $n \geq 4$, so it is better, in general, to use the Gaussian elimination or Gauss–Jordan reduction methods.

Note that at this point we have shown that the following statements are equivalent for an $n \times n$ matrix A :

1. A is nonsingular.
 2. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
 3. A is row (column) equivalent to I_n .
 4. The linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $n \times 1$ matrix $\mathbf{b}$.
 5. A is a product of elementary matrices.
 6. $\det(A) \neq 0$.
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